

SIMULATION DESIGN

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INTRODUCTION

- ❖ Simulation means an imitation of the operation of a real world situation or scenario over time.
- ❖ Also, can be defined as –
 - Re-creating the World in the Laboratory;
 - attempt to re-create a real-world situation in the laboratory closely.
- ❖ It is a replication of an environment which is under one's own control.
- ❖ Carefully designed and executed simulation increases the generality of results.
- ❖ **Example:** Suppose we want to design an electronic circuit. We do not do directly on the breadboard connecting and trying repeatedly whether it works fine because this may destroy the components. Instead we design the circuit in computer simulator first so that we test and can rectify the errors, if needed.

OUTLINE OF SIMULATION RESEARCH

A simulation study involves the following **seven generic steps** (Gilbert and Troitzsch 2005, Davis et al. 2007)

A. Research question

- ❖ Identify a research question or problem that is suitable for study by simulation.

B. Model design

- ❖ Develop schematics and conceptual diagram(s) of the system.
- ❖ Model design will usually involve some data collection to inform the parameters for the model and the initial conditions for the simulation.

C. Model building

- ❖ Transform the conceptual model into an acceptable form to a simulation software.
- ❖ A good number of software are available to support simulation studies. If none found suitable for your problem, you need to write/code the program yourself.

D. Model verification

- ❖ Verification involves running the simulation and testing whether or not the model is working as it should. If there is any problem with the simulation these should be corrected.

E. Run the simulation

- ❖ Simulations can be thought of as virtual experiments during which you run a series of 'experiments' under different conditions that can be varied as required (Davis et al. 2007).
- ❖ Harrison et al. (2007) identify **five** elements to such an experiment-
 - initial conditions,
 - time structure,
 - outcome measurement,
 - number of iterations, and
 - any variation in model parameters or initial conditions.
- ❖ Variation allows different assumptions to be tested in order to answer the research questions and also to test the sensitivity of the model to changes in parameters.

F. Model validation

- ❖ Validation helps confirm that the simulation well represents the chosen target.
- ❖ We can achieve it by comparing simulation results with empirical results.
- ❖ Validation can challenging depending on the nature of simulation and potential limitations of available empirical data.

G Findings/Inferences and Concluding Remarks

- ❖ Outcomes or findings should be specified in response to the research questions and the results should be interpreted adequately. Finally, the concluding remarks are to be given.
- ❖ Gilbert and Troitzsch (2005) note that providing enough detail for the study to be replicated while avoiding burying the reader in detail can be a particular challenge when reporting simulation research.

WHY SIMULATE ?

One can decide to carry out simulation study for a variety of reasons rather than conducting research in the real world.

- ❖ The behavior of interest could not be studied **ethically** in the real world.
- ❖ Studying a behavior under its naturally occurring conditions is **expensive**.
- ❖ Conducting research in real time may be **time consuming**.
- ❖ Control over variables; free from **extraneous or irrelevant** variables.

DECISIONS FOR SIMULATION STUDY

- ❖ Simulation may attract researches for its exciting benefits but if it is not designed and applied appropriately, the study may not produce meaningful results.
- ❖ To decide whether simulation is the correct alternative to solve a research problem, four parameters are usually critically considered :
 - Type of Problem,
 - Availability of Resources,
 - Cost involved, and
 - Availability of Data

Type of Problem

- ❖ If the problem can be solved by performing direct experiments on the system to be evaluated, then conducting direct experiments may be more desirable than simulation.
- ❖ If the system is too complex, cannot be defined, and not understandable then simulation will not produce meaningful results.
- ❖ If a problem can be solved by common sense or analytically, the use of simulation is unnecessary.
- ❖ Additionally, using algorithms and mathematical equations may be faster and less expensive than simulation study.

Availability of Resources

- ❖ **Human** and **time** are the major critical resources for conducting a simulation study.
- ❖ An experienced analyst is the most important resource since such a person has the ability and experience to determine both the model's appropriate level of detail and how to verify and validate the model.
- ❖ Without a trained simulator, the wrong model may be developed which may lead to unreliable results.
- ❖ Additionally, the allocation of time should not be so limited so as to force the simulator to take shortcuts in designing the model.
- ❖ The schedule should allow enough time for the implementation of any necessary changes and for verification and validation to take place if the results are to be meaningful.

Costs Involved

- ❖ Cost considerations is essential for each step in the simulation process. Purchase of simulation software is the only alternative, if it is not available, and computer resources are inadequate.
- ❖ If the cost exceeds the potential savings in substituting the current system, then simulation study should not be pursued.

Availability of Data

- ❖ The necessary data should be identified and located, and if the data do not exist, then the data should be gathered or generated.
- ❖ If the data do not exist and cannot be collected or generated, then continuing with the simulation study will eventually yield unreliable and useless results.
- ❖ With the inadequate data, simulation output cannot be compared to the real system's performance, which is vital for verifying and validating the model.

DESIGNING A SIMULATION

- ❖ For a simulation to improve the generality of laboratory based research, it must be properly designed.
- ❖ Observe the actual situation and study it carefully (Winkel & Sasanoff, 1970).
- ❖ Identify the crucial elements, characteristics, behaviors, functions and then try to reproduce them in the laboratory.
- ❖ The more realistic the simulation, the greater are the chances that the results will be applicable to the simulated real-world phenomenon.

The success of your simulation depends on its realism, which is of two types.

- **Mundane realism:** It is the degree to which the simulation re-creates a real-world environment. Mundane realism refers specifically to the extent to which events in an experiment are likely to occur in the real world.
- **Experimental realism:** It concerns how involved in our study, the participants become.

ELEMENTS OF EXPERIMENTAL DESIGN

The experimental design consists of five elements (Harrison et al. (2007)) :

- Initial conditions
- Time structure
- Outcome determination
- Iterations
- Variations

A. Initial conditions

- ❖ The computational model specifies how the system changes from time t to time $t+1$, but not the state of the system at time 0, so initial conditions must be specified.
- ❖ For example, in the cultural transmission simulation, initial conditions include the number of members in the organization at the beginning of the simulation and their individual enculturation scores.

B. Time structure

- ❖ The time structure sets the length of each simulation time period and the number of time periods in the simulation run.
- ❖ The length of the time period links the simulation to observation; for example, it may be desirable for turnover rates in a simulation to correspond the realistic turnover rates for an organization.
- ❖ Once the time period is determined, the number of time periods to be simulated can be set to obtain the desired total duration of the simulation run, or a rule may be established to stop the run once certain conditions (e.g., system equilibrium) are met.

C. Outcome Determination

- ❖ The outcomes of interest are often some function of the behavior of the system and need to be calculated from system variables.
- ❖ Outcomes may be calculated for each time period or only at the end of the run, depending on the simulation's purpose.
- ❖ In the cultural transmission simulation, the outcomes of interest are the mean and standard deviation of the enculturation scores of the organizational members and the number of periods it takes the system to reach equilibrium.

D. Iterations

- ❖ In stochastic models the simulation outcomes will vary somewhat from run to run, depending on the random numbers generated, so the results of one run may not be representative of the average system behavior.
- ❖ To assess average system behavior as well as variations in behavior, multiple iterations are necessary—that is, the simulation run must be repeated many times (using different random number streams) to determine the pattern of outcomes.

E. Variations

- ❖ Finally, the entire simulation process described above may be repeated with different variations. Both the parameter values and the initial conditions can be varied for two reasons.
- ❖ First, the behavior of the system under different conditions may be of interest; the examination of such differences is often a primary reason for conducting simulation experiments.
- ❖ In the cultural transmission simulation, for example, turnover rates of organizational members are varied to examine differences in system behavior under conditions of low turnover and high turnover.
- ❖ This type of variation is analogous to standard strategies for experimental design. In both simulations and laboratory experiments, the context can be controlled and manipulated to assess the effects of variations (Burton, 2003).

LIMITATIONS OF SIMULATION STUDY

- ❖ Gaining control over variables, may lose a degree of generality of results.
- ❖ Simulation is not precise. It does not yield an answer but merely provides a set of the system's responses to different operating conditions.
- ❖ Sometimes it is difficult to interpret the simulation results.
- ❖ It may take a long time to develop a good simulation model.
- ❖ Inadequate data reduces the generality of the results.

CONCLUDING REMARKS

- ❖ An overview of simulation study, its importance, merits and limitations are presented.
- ❖ Criteria used to decide whether simulation study will be suitable or not are presented.
- ❖ Elements of experimental design are elaborated.
- ❖ Limitations of simulation design are reported.

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